# WQD7009 Big Data Applications and Analytics

*Alternative Assessment*

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# Table of Contents

**1 Introduction to the Data Transformation in the Cloud..... 3**

**2 Lessons Learned..... 3**

2.1 Contents Covered..... 3

2.2 Key Learnings (focusing on data pipeline)..... 4

**Appendix..... 5**

i) Public Profile with Badge Completion..... 5

ii) Before Completion of Course..... 5

iii) After Completion of Course..... 6

# 1 Introduction to the Data Transformation in the Cloud

Data transformation is one of the subsequent and crucial phases in the cloud, situated in between the initial phase of ingestion and the final application phase. This course covered the comprehensive data transformation journey, from collection through processing, storage, analysis and activation. It emphasises the benefits of cloud-based approaches that enable scalable and automated transformation of raw data into actionable business insights through modern pipeline architectures.

# 2 Lessons Learned

## 2.1 Contents Covered

There are a total of 3 modules covered as follow:

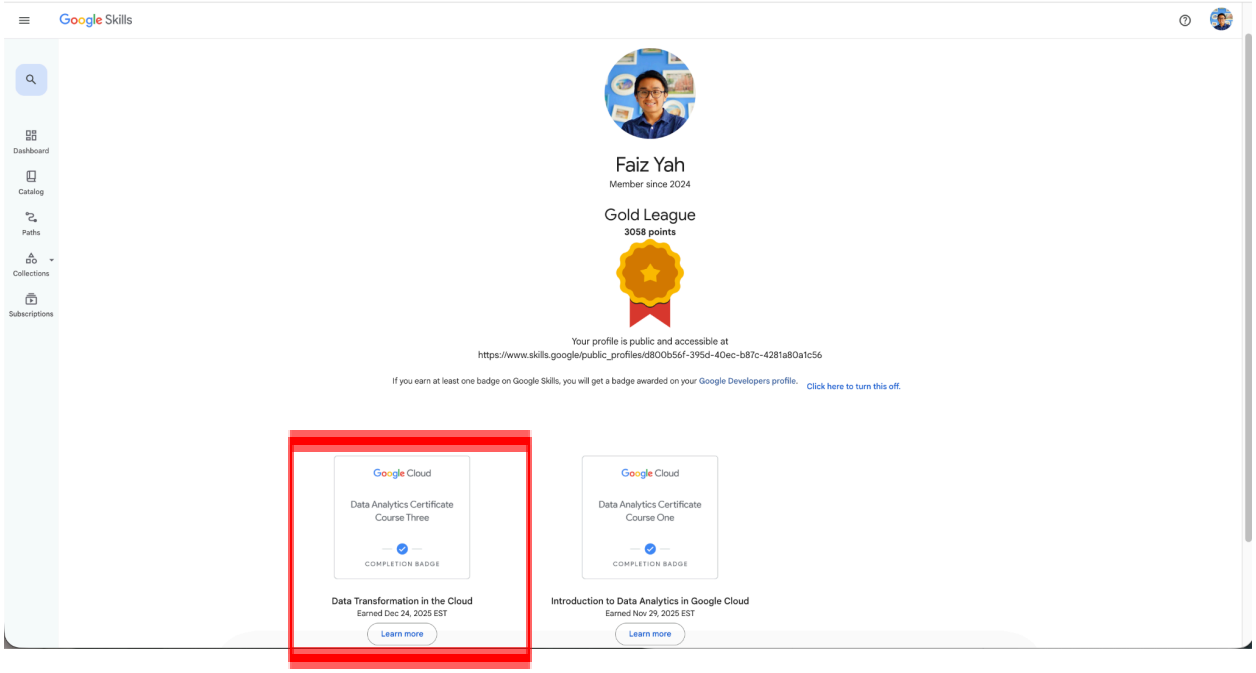
| Module | Content |
|---|---|
| **Module 1:** Introduction to data transformation in the cloud | 1. The 5 stages of data journey<br>2. Impact of data journey<br>3. Technique to transform data<br>4. Manual and automated transformation<br>5. Storage options for transformed data |
| **Module 2:** Handle raw data with data pipelines | 1. Data pipeline introduction, process and benefit<br>2. Differences between ETL and ELT<br>3. Data ingestion method and tools<br>4. Data transformation<br>5. Data validation strategy |
| **Module 3:** Cloud data optimization strategies | 1. Benefits of data transformation plan<br>2. Challenges of data transformation<br>3. Data aggregation<br>4. Duplicated data and deduplication<br>5. SQL strategies<br>6. Data derivation and manipulation |

## 2.2 Key Learnings (focusing on data pipeline)

The course contains enriching contents comprising 3 modules that explains the key concepts in data transformation. The 5 stages of data journey (collect, process, store, analyse, activate) provides a solid overview on the end-to-end phases. The distinction between ETL and ELT process demonstrated the flexibility of the cloud environment in providing options for transformation strategy. ETL (Extract, Transform, Load) has the edge in terms of providing higher data quality upfront, easier compliance and is well-established, while ELT (Extract, Load, Transform) offers faster ingestion and more importantly flexibility. Data ingestion is another critical learning point due to its importance in choosing the right option, where batch ingestion is much simpler to setup sacrificing real-time consumption, while stream ingestion is instantaneous but proves to be more complicated to manage. Next, enforcing data validation strategies such as type, format, uniqueness, correctness, null values, and duplication are also crucial in ensuring data quality. Another core takeaway is the manipulation of data, including data standardisation, enrichment, conversion, derivation and tokenisation, ensuring data is top notch in terms of quality and variety,

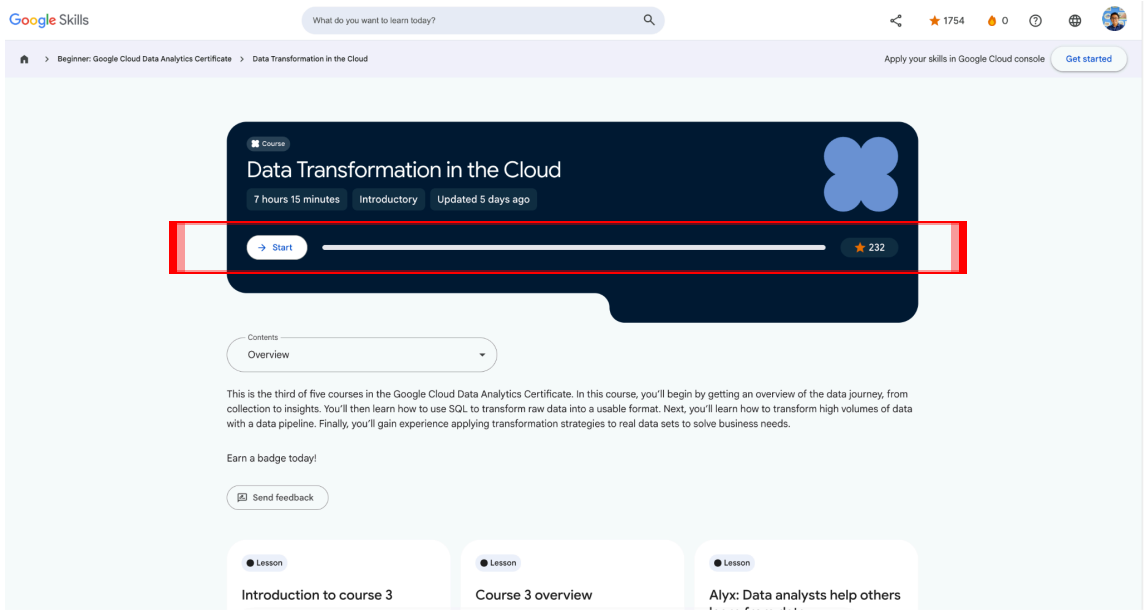
# Appendix

## i) Public Profile with Badge Completion



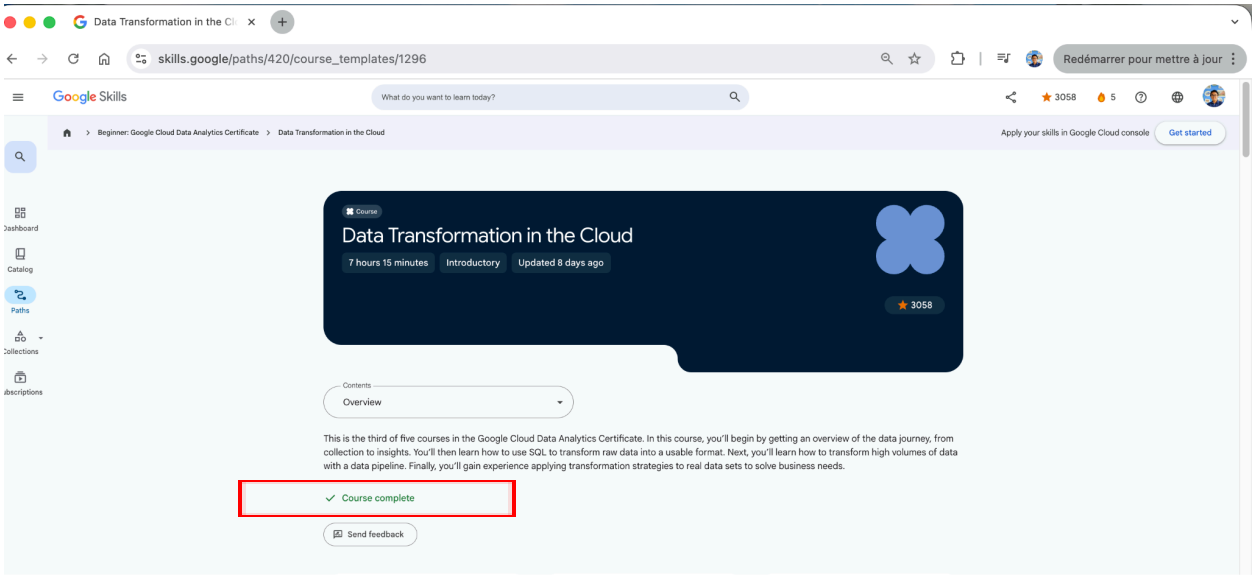

*Screenshot of Profile (link)*

## ii) Before Completion of Course




*Screenshot of Certificate Path (Before Completion of Course)*

## iii) After Completion of Course




*Screenshot of Certificate Path (After Completion of Course)*